

CHEKUNDA, Russian Federation

WMO: 315320

Lat: 50.8647N

Lon: 132.2580E

Elev: 892

StdP: 14.23

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.2	-38.9	-47.0	0.4	-41.8	-43.9	0.4	-38.4	11.9	-0.4	8.7	-11.6	0.2	0	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	85.9	68.3	82.2	66.9	78.8	65.8	71.9	79.9	70.2	77.8	68.5	75.2	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	113.2	75.1	67.9	105.8	73.0	66.2	99.9	71.7	36.4	79.6	34.9	78.2	33.4	75.2	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.3	11.1	8.3	DB	-45.1	91.2	3.5	3.2	-47.7	93.4	-49.7	95.3	-51.7	97.0	-54.3	99.3
			WB	-45.2	75.5	3.3	2.5	-47.6	77.3	-49.6	78.8	-51.4	80.2	-53.9	82.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	-21.2	-9.5	12.9	35.2	51.3	62.2	68.6	65.0	51.3	33.3	4.8	-18.8		
	DBStd	33.08	9.40	10.31	11.22	8.06	6.81	6.16	4.73	5.46	7.29	8.27	15.21	10.85		
	HDD50	9621	2206	1666	1150	449	67	2	0	1	73	519	1356	2132		
	HDD65	13704	2671	2086	1615	895	428	123	20	67	413	983	1806	2597		
	CDD50	1630	0	0	0	4	106	368	576	465	110	1	0	0		
	CDD65	240	0	0	0	0	2	40	131	66	1	0	0	0		
	CDH74	2057	0	0	0	1	86	511	1016	431	12	0	0	0		
	CDH80	548	0	0	0	0	12	137	322	76	1	0	0	0		
(14) Wind	WSAvg	2.5	1.1	1.7	2.9	3.8	3.7	2.8	2.3	2.4	2.8	3.4	2.3	1.2		
(15) Precipitation	PrecAvg	26.5	0.2	0.2	0.4	1.3	2.5	4.0	6.3	5.9	3.6	1.1	0.7	0.4		
	PrecMax	37.2	0.9	0.7	0.9	2.6	4.8	6.6	13.5	9.6	7.9	2.4	1.1	1.1		
	PrecMin	19.9	0.0	0.0	0.0	0.2	0.7	1.0	2.4	3.8	0.8	0.2	0.1	0.1		
	PrecStd	4.2	0.2	0.2	0.2	0.6	0.9	1.7	3.0	1.6	1.9	0.7	0.3	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	23.7	45.6	68.7	82.5	89.1	91.8	86.0	75.4	60.6	39.2	17.3		
		MCWB	9.9	20.0	36.3	50.4	60.1	72.0	69.8	68.1	64.3	48.2	34.7	15.0		
	2%	DB	2.2	17.3	39.7	60.4	76.1	83.8	87.6	82.1	71.3	53.8	33.8	9.1		
		MCWB	1.2	14.6	32.4	46.0	56.0	65.1	68.9	68.8	60.5	44.2	30.8	8.3		
	5%	DB	-3.1	12.1	35.2	54.7	71.5	79.8	83.6	78.8	67.5	49.5	30.5	3.0		
		MCWB	-4.0	10.1	29.4	42.3	54.2	63.7	68.2	67.2	58.0	41.9	28.4	2.2		
	10%	DB	-7.4	7.4	31.0	49.7	66.8	76.0	79.7	75.8	63.9	45.4	26.0	-3.0		
		MCWB	-8.1	5.6	26.2	39.2	51.7	62.7	67.6	65.9	56.2	39.5	24.1	-3.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	20.4	36.9	52.5	62.1	74.4	75.3	74.1	66.7	51.2	34.8	15.1		
		MCDB	11.2	23.4	43.7	66.9	75.3	84.2	83.5	80.3	73.6	57.7	37.7	16.7		
	2%	WB	1.3	14.7	33.1	47.4	58.3	69.9	72.2	71.3	62.8	46.3	31.4	8.2		
		MCDB	2.1	17.4	38.7	57.8	70.5	80.2	80.5	77.7	68.8	52.0	33.5	9.0		
	5%	WB	-3.9	10.2	29.7	43.4	56.0	66.6	70.7	69.4	60.3	43.1	28.3	2.2		
		MCDB	-3.2	12.1	35.0	52.6	66.8	75.1	78.8	75.3	65.3	47.5	30.8	3.1		
	10%	WB	-8.0	5.7	26.2	40.0	53.6	64.2	69.3	67.6	57.7	40.4	24.0	-3.6		
		MCDB	-7.3	7.5	30.9	48.8	63.9	72.2	76.7	73.0	62.3	44.4	26.1	-3.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	25.0	25.8	22.3	24.0	22.1	20.0	19.3	20.7	17.6	16.3	16.1		
		MCDBR	18.2	25.5	25.6	31.0	34.1	29.9	27.9	24.4	25.4	25.0	15.9	15.9		
		MCWBR	17.1	22.8	18.7	17.8	16.3	13.6	11.5	11.9	14.3	16.2	13.0	14.8		
	5% WB	MCDBR	18.0	25.3	24.0	28.8	28.5	25.8	21.2	19.1	21.6	20.8	15.5	15.3		
		MCWBR	17.0	22.5	17.5	17.4	15.0	13.7	11.0	10.9	13.5	14.8	12.9	14.6		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.298	0.365	0.493	0.559	0.511	0.486	0.416	0.386	0.394	0.326	0.268		
	taud		2.152	2.062	1.934	1.791	1.765	1.981	2.109	2.312	2.309	2.093	2.091	2.110		
	Ebn at Noon		238	257	257	235	226	240	245	258	253	217	203	212		
	Edh at Noon		23	34	48	62	67	54	47	37	33	34	25	21		
(44) All-Sky Solar Radiation	RadAvg		516	891	1392	1579	1576	1754	1622	1435	1096	705	487	400		
	RadStd		14	22	60	160	150	207	127	119	111	51	37	17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	+0.85	+1.10	N/A	N/A	N/A	N/A

Nomenclature: See separate page